

Mechanical Evaluation of Outsider Scientific Contributions

The LEMU Framework: Replacing Credential Filtering with Content Evaluation

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1. Definitions

This paper uses four definitions. Each is mechanical. Each is testable. Each is applied to observable structure, not to individuals. Everything in the paper is built from these definitions. A reader who does not accept them should stop here — nothing that follows will be persuasive because everything that follows depends on them.

Falsification. A claim is scientific if and only if it specifies conditions under which it would be false. This is Karl Popper's demarcation criterion, formalized in 1934. It is not merely a philosophical position. It is the structural feature that separates claims that can teach us about reality from claims that cannot. A claim that is compatible with every possible observation says nothing about which observation will actually occur. It cannot be wrong, so it cannot be tested, so it cannot teach.

Falsification as a practice requires four elements. First, commitment: the claimant states a specific prediction before the test. Second, specificity: the prediction is precise enough that observation can disagree with it. Third, priority: the prediction is stated before the result, not retrofitted. Fourth, acceptance: when observation contradicts the prediction, the claimant accepts the failure as a finding. Remove any one element and the activity is not falsification. It may be sophisticated. It may be interesting. But it is not the practice that produces scientific knowledge.

The Crank Label. In institutional science, the word “crank” is applied to people outside the credentialing system who engage with frontier scientific questions. The label has a specific structural property: it is applied based on who the person is — their degrees, their affiliation, their publication history, their institutional membership — rather than what their work contains. Once applied, the label determines the response. The work is not examined. The work is not evaluated against any content criterion. The classification intercepts the work before examination can occur.

This classification is the structural inverse of falsification. Falsification evaluates claims on their content — their specificity, their testability, their agreement with observation. The crank label evaluates claimants on their credentials — their institutional history, their social position within the field. Falsification asks “what does the work say and can it be tested?” The crank label asks “who is the author and are they authorized to speak?” These are opposite operations. The label structurally prevents falsification from being applied to the work it intercepts.

LEMU. This paper proposes four evaluation axes that replace credential-based filtering with content-based evaluation. The axes are Logic, Empirical, Mathematical, and Utility. Each axis is non-subjective. Each produces a result that can be explained mechanically to the submitter. Each evaluates the work, not the author.

The axes are defined in full in Section 3. The critical preliminary definition here is the distinction between utility and novelty, because the two will be confused if not separated at the outset. Novelty asks whether something has been said before. Utility asks whether it does anything. These are different properties. A paper that independently re-derives a known result using a new method has low novelty but potentially high utility — the new method might be simpler, faster, or applicable to problems the existing method cannot reach. A paper that proposes something never proposed before but with no consequences has high novelty and zero utility. This paper uses only utility. Novelty is excluded on purpose. Whether a contribution is original is a separate question from whether it is valuable. The evaluation framework does not ask “is this new?” It asks “does this work?”

Located Error. When a specific claim fails at a specific point for a specific identifiable reason, the failure is a located error. Located errors are the most valuable output of any evaluation process because they narrow the search space. Before the evaluation, the problem could be anywhere in the work. After a located error, the problem is at a specific point. The submitter knows what to fix. The evaluator knows what failed. The search space has been reduced by the precise amount of information contained in the located error. A rejection that says “this work does not meet our standards” is not a located error. It provides no information about where the work fails. It does not narrow the search space. It cannot be acted on.

2. The Current System and What It Produces

A person outside the institutional credentialing system — someone without the expected degrees, affiliations, and publication history — submits work to a scientific journal or

presents it to the scientific community. The current evaluation process operates as follows.

The first evaluation is credential-based. Does the author hold a doctorate in the relevant field? Are they affiliated with a recognized institution? Have they published in peer-reviewed journals? Do they use the field's standard vocabulary and citation conventions? These questions are about the author. They tell you about the author's institutional history. They tell you nothing about whether the author's current work is internally consistent, empirically grounded, mathematically correct, or useful.

If the author does not pass the credential check, the work is classified as crank and excluded. The classification happens before the content is examined. Whether the work contains valid logic, references real measurements, performs correct mathematics, or would be useful if true — none of these questions are asked. The classification was applied before the questions could be reached.

The submitter receives a rejection. The rejection does not explain where the work fails in terms of content. It does not identify logical errors. It does not point to empirical disconnects between the work's claims and established measurements. It does not locate mathematical mistakes. It does not assess whether the work's conclusions, if true, would matter. It provides no located error. It provides no information the submitter can act on.

The submitter cannot improve because they have received no information about what specifically is wrong with their work. They believe their work has merit — and they may be right or wrong about this, but they cannot determine which because no content evaluation was performed. They revise based on guesswork or they resubmit unchanged. The institution receives another submission. The submission is classified as crank again. The cycle reinforces itself. Each iteration confirms the institution's belief that outsider submissions are uniformly worthless. Each iteration confirms the outsider's belief that the institution is not evaluating their work. Both beliefs are self-reinforcing. Both may be wrong. Neither can be tested because the system never reaches the content.

The cost to the institution is invisible by construction. The institution cannot know what it discards because it never examines what it discards. The discarded pool contains work that ranges from incoherent to potentially valuable. The institution treats the entire range as a single category. A perpetual motion proposal and a reproducible derivation matching established measurements to sub-percent precision receive the same classification, the same response, and the same exclusion.

The cost to the submitter is visible and concrete. Time invested in work that will never be read. Effort spent meeting standards that are not the actual evaluation criteria. The experience of being excluded based on identity rather than content, by an institution that claims to evaluate ideas on their merits.

The cost to the field is structural. People with non-standard backgrounds — systems engineers, software architects, information theorists, applied mathematicians — bring categories of thinking that the field's training pipeline does not produce. An engineer who practices daily commitment to specific values with explicit failure conditions brings Popper's epistemology in its practiced form. A software architect who thinks in terms of discrete computation, finite state, and deterministic update brings a structural perspective

that continuous-mathematics training does not generate. These contributions are excluded before examination because the contributors do not hold the expected credentials. The field loses entire categories of methodology it cannot generate internally and does not evaluate externally.

3. The LEMU Evaluation Axes

Four axes. Each is defined mechanically. Each has explicit evaluation criteria. Each produces a result that can be explained to the submitter with specific reference to their work. Each evaluates the work on its content. The author's identity, credentials, affiliation, and institutional history are not part of any axis.

3.1 Logic

Is the paper internally consistent? Do the conclusions follow from the premises? Can the logical structure be traced from assumptions through argument to conclusions without a break?

The evaluation is mechanical. The reviewer identifies the paper's stated premises — the assumptions it declares as starting points. The reviewer traces the argument step by step. At each step, the reviewer asks: does this follow from what precedes it? Is there a logical gap — a step that requires an assumption not stated in the premises? Is there a contradiction — a step that asserts something incompatible with an earlier step? Does the conclusion follow from the chain?

A pass looks like this: premises are stated explicitly. Each step follows from the previous by identifiable reasoning. No step contradicts an earlier step. The conclusion follows from the chain. The reader can reconstruct the argument independently by following the stated steps.

A failure looks like this: the paper assumes a quantity is conserved in paragraph three and treats it as non-conserved in paragraph seven. Or the conclusion requires a step that is never provided — there is a gap in the chain where the reader cannot get from one claim to the next without adding an unstated assumption. Or the paper contains a circular argument — the conclusion is used as a premise for its own derivation.

The logical evaluation does not require deep domain expertise. It requires careful reading and the ability to follow a chain of reasoning. An undergraduate trained in formal logic or mathematical proof can perform this evaluation. The evaluation produces a specific result: the logic is consistent, or it fails at an identified point for an identified reason. The identified point and reason are the located error provided to the submitter.

3.2 Empirical

Does the paper connect to physical reality through measured values? Are the values it uses actual measurements from established sources? Does it make predictions that can be compared against observation?

The evaluation is mechanical. The reviewer identifies every physical value the paper uses. For each value, the reviewer checks: is this a real measured quantity? Is the value the paper uses consistent with the established measurement? If the paper derives predictions, do those predictions correspond to quantities that have been measured or could be measured?

A pass looks like this: the paper uses the measured value of the electron mass as an input. It derives a prediction for the muon mass. The predicted value can be compared against the measured muon mass. The paper states the comparison explicitly and provides the measured value alongside its prediction. The agreement or disagreement is quantified.

A failure looks like this: the paper uses values that do not correspond to any established measurement. Or it references real quantities but uses incorrect values. Or it makes predictions that cannot be compared against any observable quantity — predictions about things that have not been measured and for which no measurement procedure exists. Or it makes predictions but does not compare them against available measurements when those measurements exist.

This evaluation requires access to reference data — standard compilations of physical measurements. It does not require the evaluator to have performed the measurements or to hold expertise in the measurement techniques. It requires looking up values and comparing them. An undergraduate with access to standard reference data can perform this evaluation.

3.3 Mathematical

Does the math work? Do the equations follow from the stated premises? Are the derivations correct step by step?

The evaluation is mechanical. The reviewer follows each derivation from its starting equation to its result. At each step, the reviewer checks: is the algebraic manipulation correct? Are the units consistent? Is any approximation stated and justified? Does the final result follow from the chain of steps?

A pass looks like this: each equation follows from the previous by identifiable mathematical operations. Units are consistent throughout. Approximations are stated with their conditions of validity. The final result is correctly computed from the inputs. An independent person performing the same derivation from the same starting point arrives at the same result.

A failure looks like this: a sign error at a specific step. A term dropped between one equation and the next. An unjustified step — a manipulation that does not follow from standard mathematical operations. A dimensional inconsistency — quantities with incompatible units being added or equated. A division by a quantity that could be zero without treatment of that case. Each of these is a located error at a specific point in a specific derivation.

This evaluation requires mathematical training appropriate to the level of the mathematics used in the paper. For papers using algebra, an undergraduate can perform it. For papers

using advanced techniques, a graduate student with the relevant mathematical background performs it. The evaluation is not subjective. A derivation is either correct or it contains an identifiable error at an identifiable step.

3.4 Utility

If the paper's claims are true, does that matter?

This is the axis most likely to be misapplied, so the criteria must be precise. Utility is not novelty. Utility is not prestige. Utility is not alignment with current research priorities. Utility is: does the work do something?

Utility has four sub-criteria, any one of which is sufficient. First, the work proves something — it establishes a result that has consequences for other calculations, other measurements, or other theoretical frameworks. Second, the work can be used for something — it provides a method, a value, a framework, or a tool that enables work that was not previously possible or was previously more difficult. Third, the work means something — it reveals a connection, a pattern, or a structural relationship that was not previously visible. Fourth, the work closes something — it answers a question definitively, either positively or negatively, that was previously open.

A pass looks like this: the paper derives a physical constant from declared integer inputs. If the derivation is correct and the result matches measurement, then the constant can be computed rather than only measured. That is utility — it means something about the relationship between the constant and the inputs, and it can be used as an independent check on measurement.

A failure looks like this: the paper proves that a mathematical identity holds. The identity is correct. It is also inert — it does not connect to any physical quantity, does not enable any calculation that was not previously possible, does not reveal any relationship that was not previously known, and does not close any open question. The statement “cat = cat” is logically valid, empirically meaningless, mathematically trivial, and has zero utility. It is true and it does nothing.

The utility evaluation requires the reviewer to assess consequences. This is the axis where domain knowledge is most relevant and where the evaluation is most likely to vary between reviewers. However, the sub-criteria are concrete. The reviewer can ask: if this result is correct, what else follows from it? If nothing follows, utility is zero. If specific consequences follow, the reviewer identifies them. The submitter receives an explanation of what their work does or does not accomplish in terms of consequences for the field.

4. The Triage Pipeline

Three levels. Each level has a defined role, defined criteria for escalation, and defined output to the submitter. The credential status of the author is not an input at any level.

4.1 Level 1: Undergraduate Review

The undergraduate evaluates the paper against all four LEMU axes. This is the entry point for every submission regardless of the author's background.

The undergraduate has been trained in the LEMU evaluation criteria. They have access to standard reference data for the Empirical axis. They have the mathematical background to check derivations at the level of the paper's mathematics, or the paper is routed to an undergraduate with the appropriate mathematical training. They produce a LEMU scorecard: each axis rated with a specific explanation referencing the paper's content.

The scorecard is not a single number. It is four assessments, each with located findings. "Logic: consistent. No internal contradictions identified. The argument from premise A through steps B, C, D to conclusion E is traceable." Or: "Logic: fails at step C. Step C requires conservation of angular momentum, which contradicts the paper's assumption in premise A that the system is open to external torques." Each assessment points at specific content in the paper.

If the paper scores zero on three or more axes — meaning it is logically inconsistent, empirically ungrounded, mathematically incorrect, or without utility on at least three of the four dimensions — it is returned to the submitter with the full scorecard. The submitter now has a mechanical explanation of where their work fails, referencing specific points in their paper. They know what to fix. This is the located error the current system never provides.

If the paper scores above zero on at least three axes, it escalates to Level 2.

4.2 Level 2: Graduate Student Review

The graduate student brings domain expertise the undergraduate does not have. They can assess whether the empirical claims match current measurements at the stated precision and whether the paper accounts for known experimental uncertainties. They can evaluate whether the mathematical framework is appropriate for the physical system being described — not just whether the math is correct internally but whether it is the right math for the problem. They can evaluate utility with deeper knowledge of the field's open questions and existing approaches.

The graduate student produces an expanded evaluation. The scorecard is refined with domain-specific context. "Empirical: the paper's predicted value for the neutron magnetic moment is within 0.2% of the measured value. However, the paper does not account for radiative corrections, which are known to contribute at the 0.1% level. The prediction should be compared against the corrected value." This is specific, actionable, and located.

If the paper fails at this level, the submitter receives the expanded evaluation. If it passes, it escalates to Level 3.

4.3 Level 3: Faculty Review

The paper has survived two levels of content-based evaluation. The faculty member is evaluating work, not a person. The credential question has been mechanically removed

by the pipeline — the paper arrived here because of what it contains. The faculty member did not choose to read it based on who wrote it. They read it because it passed two independent content evaluations.

The faculty member makes the determination: does this work merit engagement by the field? The options are concrete. Publication in a journal. Experimental test of the paper's predictions. Collaboration with the author. Integration with existing research programs. Or: the work passes LEMU evaluation but does not rise to the level of field engagement, with a specific explanation of why.

At every level, the output to the submitter is a mechanical explanation referencing LEMU. The submitter always knows where they stand and why. The explanation references their work's content, not their credentials. The process is the same regardless of whether the author holds a doctorate from a leading university or has no formal physics education at all.

5. The Pedagogical Feedback Loop

The LEMU pipeline is not only a filter. It is a data source. This section describes what the data contains and what it reveals.

Undergraduates and graduate students processing outsider papers against LEMU criteria are observing the actual failure modes of reasoning about physics. Not textbook examples of errors constructed for teaching purposes — real errors made by real people who were trying to solve real problems with the physics knowledge they had available.

These errors have patterns. If fifty papers in a year all misapply conservation of energy in the same structural way — say, treating kinetic energy as conserved in inelastic collisions — that is not fifty independent mistakes. It is a signal. The signal tells you that somewhere in the educational pipeline, conservation of energy is being taught in a way that produces that specific misapplication. Perhaps the distinction between elastic and inelastic collisions is taught as a classification exercise rather than as a consequence of the underlying force dynamics. Perhaps the examples used in teaching all happen to be elastic, so students generalize incorrectly. The pattern in the errors points at the upstream cause.

The institution controls physics education at every level. It sets university curricula. Its members write textbooks. Its members train high school teachers. Its members influence national curriculum standards. The institution then receives outsider submissions that reflect the education it provided — or failed to provide. The errors in those submissions are the institution's own output, reflected back. They are direct evidence of where the educational pipeline fails to deliver correct understanding.

Examining these patterns is examining the institution's own effectiveness as an educator. The information is generated for free as a byproduct of the evaluation process. It requires no additional experiment, no additional funding, no additional infrastructure. It is already

there in the submissions. The current system throws it away because it throws away the submissions without reading them.

The feedback loop, when operating, functions as follows. Education produces people with a specific understanding of physics. Those people produce work that reflects their understanding. The work is evaluated against LEMU. The evaluation reveals patterns in how physics is misunderstood outside the institution. The patterns identify specific points where education fails to deliver correct understanding. Those points inform targeted improvements to the educational pipeline. The improvements change what future students learn. Future students produce work with different error patterns. The new patterns reveal the next set of educational gaps. Each cycle is an iteration that improves the next.

This is the same iterative improvement process that every engineering discipline uses. Software organizations study their bug reports to identify systematic problems in their development process. Manufacturing facilities study their defect rates to identify systematic problems in their production lines. The physics institution has access to the equivalent data — systematic patterns in how people fail to understand physics — and currently discards it without examination.

The benefit to the undergraduate and graduate student reviewers is direct and substantial. Processing real papers against LEMU criteria teaches evaluation skills that no coursework provides. The reviewer learns to identify logical gaps, empirical disconnects, mathematical errors, and utility failures in real work produced by real people with real intentions. These skills transfer directly to evaluating their own work, their colleagues' work, and published literature. A graduate student who has processed fifty outsider papers through LEMU evaluation has fifty experiences of locating specific errors in specific arguments. That is more evaluation practice than most graduate programs provide in their entirety.

6. The Data Validity Argument

Every empirical field that studies human behavior or human-produced output treats non-expert data as a valid source of structural information.

Anthropologists study cultural artifacts produced by people who are not anthropologists. The artifacts contain the information the anthropologist seeks precisely because they were produced by people engaged in the activity being studied, not by experts performing the activity for academic purposes. Sociologists study survey responses from people who are not sociologists. The responses are the data. Their value lies in what they reveal about the respondents' actual beliefs and behaviors, not in their methodological sophistication. Epidemiologists study patient reports from people who are not epidemiologists. The reports contain errors, self-diagnosis, and misunderstanding. They also contain patterns that reveal the spread, symptoms, and risk factors of disease. Linguists study informal speech produced by people who are not linguists. The speech patterns — including the errors — are the data that reveals how language actually functions in use.

In every case, the raw data is produced by non-experts. In every case, it contains errors. In every case, the errors themselves are informative — they reveal patterns about how non-

experts understand, misunderstand, and engage with the domain. In every case, the field treats the data as worth examining because the structure in the data is valuable regardless of the credentials of the person who produced it.

Outsider physics papers are data about how people think about physics. They are produced by people who are engaged with physics questions — people who have invested time and effort in trying to understand fundamental physical problems. The papers contain errors. They also contain structural patterns about how physics is understood outside the institution, novel framings that arise from non-standard training backgrounds, cross-domain methodologies imported from fields that practice different forms of rigor, and occasionally correct results that the institution has not produced.

The physics community is the one empirical field that refuses to examine its most abundant source of non-expert data about its own domain. Every other field that studies human-produced data recognizes that the data's value is in its structure, not in the credentials of its producer. Physics discards the data unexamined on the basis of who produced it. This is a status judgment — outsider work is low-prestige, so it is treated as low-value. It is not a scientific judgment. A scientific judgment would be: examine the data, characterize its structure, extract what is informative, and discard what is not. That is what LEMU evaluation does.

7. The Transparency Demand

The current institutional position on outsider contributions exists in a state of practiced ambiguity. Journals publish submission guidelines that do not mention credential requirements. The review process is described in terms of scientific merit and methodological rigor. The stated position is that anyone can submit work and the work will be evaluated on its content.

The practiced position is different. Uncredentialed authors face barriers that credentialed authors do not. Submissions are held longer. Rejections are more frequent and less detailed. The crank classification is applied by reviewers and editors based on the author's identity before the content is examined. The stated openness and the practiced exclusion coexist in the same system.

This ambiguity is the worst possible state for everyone involved.

It is worse than genuine openness because it filters on credentials while claiming to filter on content, which means the filtering cannot be challenged — a submitter who suspects credential bias is told the rejection was based on scientific merit, and they have no way to verify this because the evaluation was not mechanical and produced no LEMU-style scorecard.

It is worse than stated closure because it wastes the time and effort of submitters who believe the stated openness is real. People invest years of work on the assumption that their submission will be evaluated on its content. The discovery that this assumption was wrong comes only after the investment has been made. A stated closure would be honest.

People would know before investing their effort that the gate is closed. They could make informed decisions about where to direct their work.

The ambiguity also costs the institution. It generates repeated submissions from people who received no located error and therefore cannot improve. It generates hostility from people who discover the gap between stated policy and practiced behavior. It generates a growing population of people who correctly perceive that the institution's evaluation process is not what it claims to be, which damages the institution's credibility on the separate and more important question of whether its scientific claims are reliable.

This paper demands a resolution. The institution has three options. First: implement a genuine content-based evaluation process such as the LEMU pipeline described in this paper, where every submission regardless of author credentials is evaluated mechanically against defined content criteria and receives a located explanation of the result. Second: state publicly, formally, and without ambiguity that submissions from authors without specified credentials are not accepted, so that no one wastes time and effort on work that will never be examined. Third: continue the current ambiguity, which costs everyone and benefits no one.

The first option is the only one consistent with falsification as an epistemological foundation. A field that claims to evaluate claims on their content, that claims falsification as its core methodology, that claims openness to evidence regardless of source — that field must evaluate outsider claims on their content, or it must stop claiming that it does. The LEMU pipeline is one implementation. Other implementations with different criteria and different structures could serve the same function. The specific implementation matters less than the structural requirement: evaluate work on content, provide located errors, and do not filter on credentials.

8. Falsification Conditions

This paper practices what it prescribes. Every major claim has an explicit failure condition. Each failure condition is observable. If the claim fails, the failure is this paper's contribution — a located error identifying where the structural analysis is wrong.

Claim 1: LEMU evaluation is non-subjective and produces consistent results. The test: present the same paper to multiple independent evaluators trained in the LEMU criteria. Each evaluator produces a scorecard independently. If evaluators produce substantially different scores on the same paper using the same criteria — if one evaluator finds the logic consistent and another finds a logical failure at a specific step, and both are applying the criteria correctly — then the criteria are not sufficiently mechanical. The evaluation framework fails and requires revision to achieve consistency.

Claim 2: The triage pipeline surfaces work that credential-based filtering would exclude. The test: run both evaluation systems in parallel on the same pool of submissions. The credential system classifies as it currently does. The LEMU pipeline evaluates on content. If the LEMU pipeline does not surface any work to Level 3 that the credential system would have excluded — if every paper that passes LEMU also passes credential

filtering, and every paper that fails credential filtering also fails LEMU — then the credential system is not excluding valuable work and the LEMU pipeline adds no evaluation value beyond the current system. The pipeline would retain value for providing located errors to submitters but its primary claim about recovering lost contributions would have failed.

Claim 3: The pedagogical feedback loop produces actionable educational information. The test: process outsider submissions through LEMU evaluation and track error patterns across the submission pool. If the errors show no identifiable patterns — if each paper fails in a unique way unrelated to other papers' failures — then the submission pool does not contain systematic pedagogical signal and the feedback loop does not function as described. Further: if educational changes informed by the observed patterns do not reduce the rate of those specific patterns in subsequent submission cycles, then the feedback loop does not produce effective educational improvements.

Claim 4: Outsider submissions contain recoverable scientific value at rates above zero. The test: process a sufficiently large representative sample of outsider submissions through the full LEMU pipeline. Track how many papers reach each level. If zero papers from a sufficiently large sample survive to Level 3, then the outsider contribution pool does not contain work that merits institutional engagement at the quality level the pipeline is designed to detect. This would not invalidate the pipeline's other functions — providing located errors and generating pedagogical data — but it would falsify the claim that the credential system is excluding scientifically valuable work.

Claim 5: The LEMU pipeline is more efficient than undifferentiated exclusion. The test: measure the total institutional resources consumed by each system — the current credential-based filtering and the LEMU pipeline — per unit of valuable work identified. If the LEMU pipeline consumes more resources per identified valuable contribution than the current system, then it is less efficient even if it recovers additional contributions. The pipeline's value would then depend on whether the recovered contributions are worth the additional cost, which is a separate assessment.

Each of these tests is specific. Each is observable. Each could falsify the corresponding claim. The paper commits to each claim and accepts falsification if it comes.

9. Closing

The institution that claims falsification as its epistemological foundation does not apply falsification to its own evaluation process. Work is accepted or rejected based on the credentials of the author. The criteria are about the person, not the work. The evaluation produces no located error. The submitter receives no information about where their work succeeds or fails. The institution receives no information about what it is discarding. The educational pipeline that produced the submitter's understanding receives no feedback about where it failed. Every party loses. No party benefits except in the narrow sense that the institution avoids the effort of reading work from people it has not credentialed.

The LEMU framework replaces this with mechanical content evaluation. Four axes —

Logic, Empirical, Mathematical, Utility — each defined with explicit criteria, each producing a located result that can be explained to the submitter. A triage pipeline that escalates work based on content scores, removing the credential question mechanically by never introducing it. A pedagogical feedback loop that mines the error patterns in submissions for information about where physics education fails. A data-validity recognition that outsider submissions are the field’s most abundant and least examined source of information about how physics is understood outside the institution.

The framework demands a transparency resolution. Evaluate outsider work on content, or state publicly that outsider work is not accepted. The current ambiguity — nominal openness with practiced exclusion — is the one option that is indefensible. It wastes resources, destroys information, damages credibility, and is inconsistent with the epistemological principles the institution claims to uphold.

The universe evaluates every interaction on its content. A photon arriving at an electron does not present credentials. The interaction proceeds based on the physical properties of the participants — energy, momentum, spin, charge. The outcome is determined by what the interaction contains, not by where the participants originated. A scientific institution that studies this universe could adopt the same evaluative principle toward the people who want to help study it.

References

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- [3] CODATA Task Group on Fundamental Constants, “CODATA Recommended Values of the Fundamental Physical Constants,” *Reviews of Modern Physics*, 2021.

Series cross-references (for deeper treatment of concepts introduced in this paper):

- Statistical breadth substituting for functional depth: [\[HOWL-CULT-1-2026\]](#)
- The institutional gap between departments: [\[HOWL-CULT-2-2026\]](#)
- The structural cessation of falsification practice: [\[HOWL-CULT-3-2026\]](#)
- What falsification structurally requires: [\[HOWL-CULT-4-2026\]](#)
- Why located errors are the most valuable finding: [\[HOWL-CULT-5-2026\]](#)
- Publications as immutable timestamps: [\[HOWL-CULT-6-2026\]](#)
- Inherited enforcement of untested norms: [\[HOWL-CULT-7-2026\]](#)
- Working in the space between departments: [\[HOWL-CULT-8-2026\]](#)
- Institutional structure for cross-domain work: [\[HOWL-CULT-9-2026\]](#)

- The structural mechanics of institutional non-commitment: [HOWL-CULT-12-2026]
- Replacing averaged snapshots with synchronized measurement: [HOWL-CULT-13-2026]

[HOWL-CULT-14-2026] Mechanical Evaluation of Outsider Scientific Contributions. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-14-2026>

[HOWL-CULT-1-2026] The N=1000 Fallacy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-1-2026>

[HOWL-CULT-2-2026] The Domain Boundary. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-2-2026>

[HOWL-CULT-3-2026] The Falsification Forfeiture. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-3-2026>

[HOWL-CULT-4-2026] What Falsification Actually Requires. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-4-2026>

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